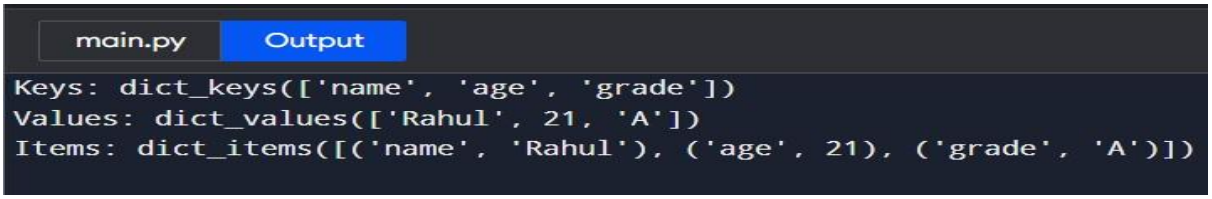
Experiment no:-3

Title:- : Write programs to implement operations like adding, updating, and deleting items in dictionaries and performing set operations (union, intersection, etc.).

1. Create a dictionary with a student's name, age, and course. Add a new key grade, update age, delete course, and print all keys, values, and items student = {"name": "Rahul", "age": 20, "course": "BCA"} student["grade"] = "A" # Add new key student["age"] = 21 # Update age del student["course"] # Delete key print("Keys:", student.keys()) print("Values:", student.values()) print("Items:", student.items())



2.Create a dictionary with 5 items. Add two new key-value pairs, update two existing values, delete one key using del and one using pop(), and iterate through keys and values

```
data = {"a": 1, "b": 2, "c": 3, "d": 4, "e":
5} data["f"] = 6 data["g"] = 7 data["a"] =
10 data["c"] = 30 del data["b"]
data.pop("e")

for key, value in data.items():
print(key, ":", value)
```



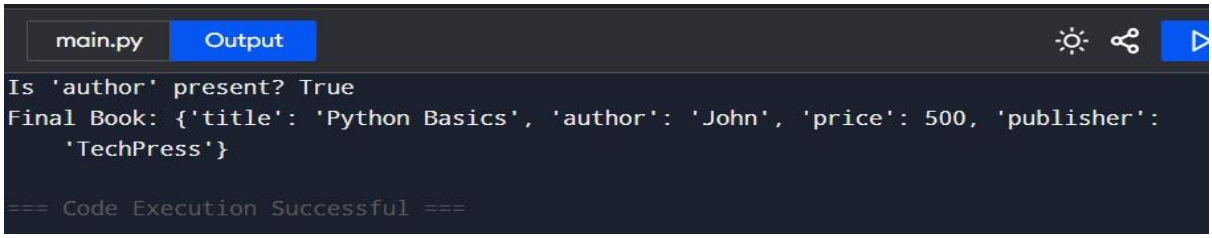
. 3. Create a dictionary to store a book's title, author, price, and pages. Add a new key publisher, update the price, delete pages, check if author exists, and print the dictionary before and after each operation.

```
book = {"title": "Python Basics", "author": "John", "price": 450, "pages": 300}

book["publisher"] = "TechPress" book["price"] = 500 del book["pages"]

print("Is 'author' present?", "author" in book)

print("Final Book:", book)
```



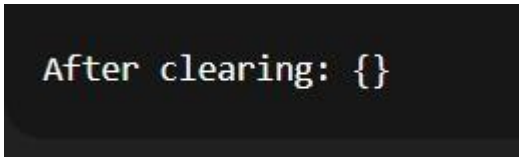
4. Create a dictionary with employee details: id, name, department, and salary. Add a new key bonus, update salary, delete department, clear all items, and print dictionary after each operation.

employee = {"id": 101, "name": "Amit", "department": "HR", "salary": 30000}

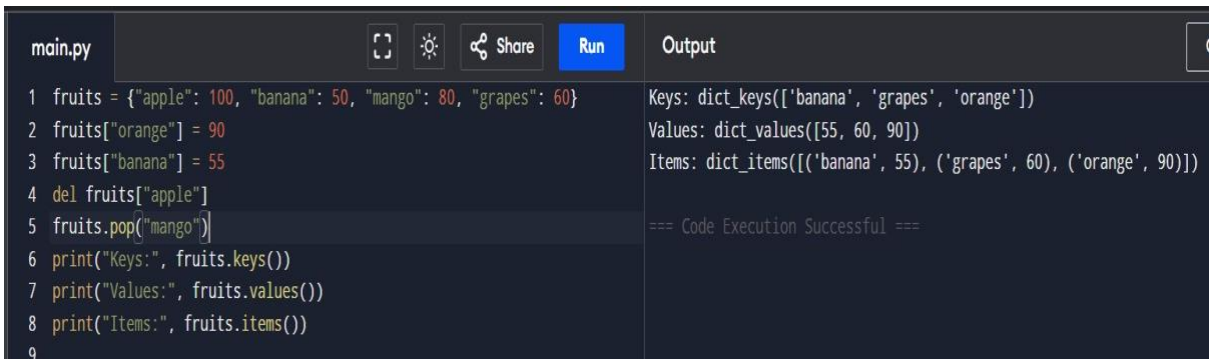
employee["bonus"] = 5000 employee["salary"] = 35000 del

employee["department"] employee.clear()

print("After clearing:", employee)



5. Create a dictionary of 4 fruits and their prices. Add a new fruit, update the price of an existing fruit, delete one fruit using del, pop another fruit, and print keys, values, and items



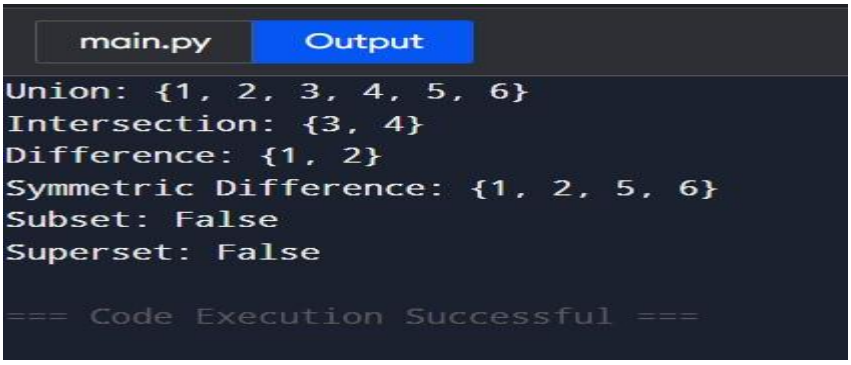
6. Create two sets of integers. A = {1, 2, 3, 4} B = {3, 4, 5, 6} print("Union:", A | B)

print("Intersection:", A & B) print("Difference:", A - B) print("Symmetric Difference:", A ^ B)

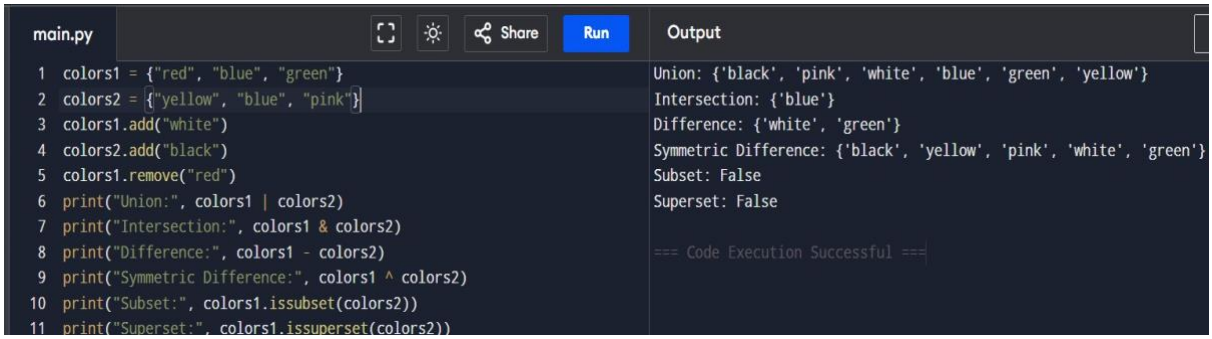
A.add(7) A.remove(2)

print("Subset:", A.issubset(B)) print("Superset:",

A.issuperset(B))



7. Create two sets of colors. Add a new color to each set, remove one color, find union, intersection, difference, symmetric difference, and check if one set is subset or superset of the other



8. Create a set of 10 numbers. Add three new numbers, remove two numbers, print the set, and perform union, intersection, difference, and symmetric difference with another set.



9. Create a dictionary with product details: name, quantity, price, and category. Add a discount key, update quantity and price, delete category, print keys, values, and items, and check if name exists.

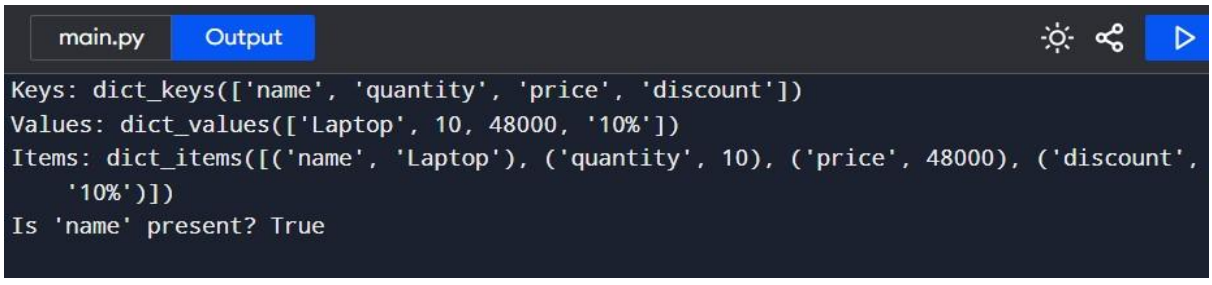
product = {"name": "Laptop", "quantity": 5, "price": 50000, "category": "Electronics"}

product["discount"] = "10%" product["quantity"] = 10 product["price"] = 48000 del

product["category"] print("Keys:", product.keys()) print("Values:", product.values())

print("Items:", product.items())

print("Is 'name' present?", "name" in product)



10. Create a dictionary with 5 student names and their scores. Add scores for 2 more students, update 3 scores, delete 1 student using del and another using pop(), and iterate through keys and values.

```
scores = {"Aman":85, "Rita":90, "Karan":70, "Meena":75, "Tina":80}

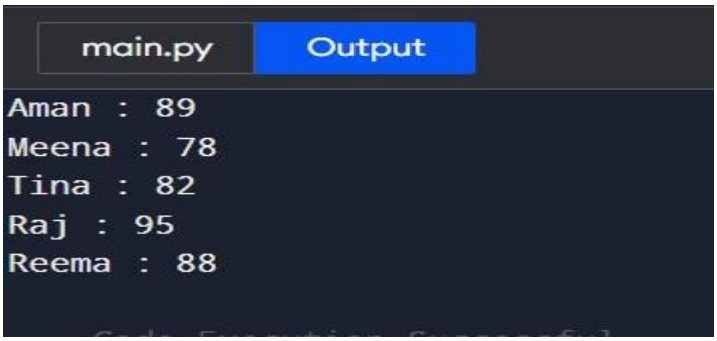
scores["Raj"] = 95 scores["Reema"] = 88 scores["Aman"] = 89

scores["Tina"] = 82 scores["Meena"] = 78 del scores["Rita"]

scores.pop("Karan")

for k, v in scores.items():

print(k, ":", v)
```



11. Create two sets of numbers representing exam scores. Perform union, intersection, difference, symmetric difference, add a new score, remove a score, and check if one set is subset or superset of the other.

```
S1 = {45,56,67,78} S2 = {56,67,88,99}

print("Union:", S1 | S2)

print("Intersection:", S1 & S2)

print("Difference:", S1 - S2)

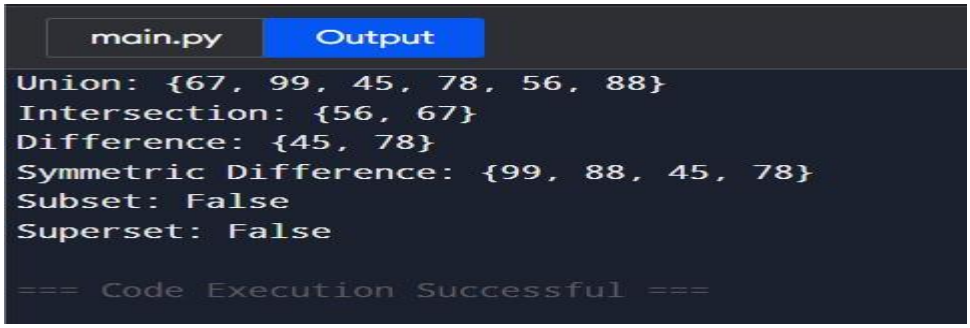
print("Symmetric Difference:", S1 ^ S2)

S1.add(100)

S1.remove(45) print("Subset:",

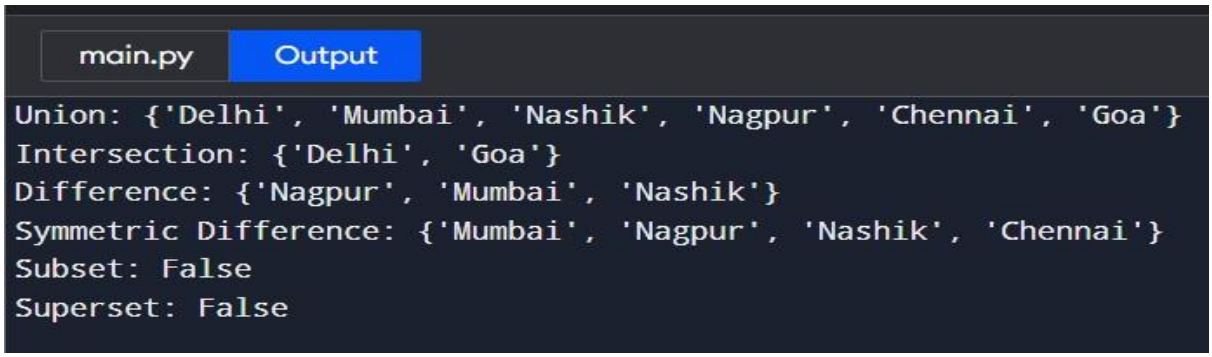
S1.issubset(S2))

print("Superset:", S1.issuperset(S2))
```

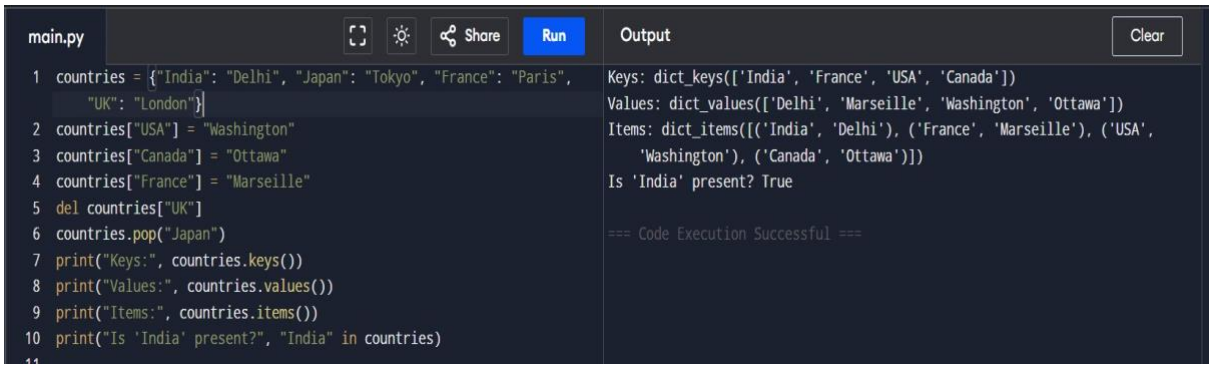


12. Create a set of cities you have visited. Add 2 new cities, remove 1 city, perform union, intersection, difference, symmetric difference with another set of cities, and check subset/superset relationships.

```
cities1 = {"Pune", "Mumbai", "Delhi", "Goa"} cities2
= {"Goa", "Chennai", "Delhi"} cities1.add("Nagpur")
cities1.add("Nashik") cities1.remove("Pune")
print("Union:", cities1 | cities2) print("Intersection:",
cities1 & cities2) print("Difference:", cities1 - cities2)
print("Symmetric Difference:", cities1 ^ cities2)
print("Subset:", cities1.issubset(cities2))
print("Superset:", cities1.issuperset(cities2))
```



13. Create a dictionary of 4 countries and their capitals. Add two more countries, update a capital, delete one country using del and another using pop(), print all keys, values, items, and check if a country exists.



14. Create a set of 8 favorite movies. Add 2 movies, remove 1 movie, perform union, intersection, difference, symmetric difference with another set, and check subset and superset.

main.py | Share | Run | Output | Clear

```
movies1 = {"Inception", "Avatar", "Titanic", "Joker", "Dune", "Matrix"
    , "RRR", "Kantara"}
movies2 = {"Avatar", "Interstellar", "RRR", "Pushpa"}
movies1.add("KGF")
movies1.add("Leo")
movies1.remove("Joker")
print("Union:", movies1 | movies2)
print("Intersection:", movies1 & movies2)
print("Difference:", movies1 - movies2)
print("Symmetric Difference:", movies1 ^ movies2)
print("Subset:", movies1.issubset(movies2))
print("Superset:", movies1.issuperset(movies2))
```

```
Union: {'Kantara', 'Matrix', 'RRR', 'Avatar', 'Interstellar', 'Inception',
    'KGF', 'Titanic', 'Dune', 'Leo', 'Pushpa'}
Intersection: {'Avatar', 'RRR'}
Difference: {'Kantara', 'Matrix', 'Inception', 'KGF', 'Titanic', 'Dune',
    'Leo'}
Symmetric Difference: {'Kantara', 'Matrix', 'Dune', 'Interstellar',
    'Inception', 'KGF', 'Titanic', 'Leo', 'Pushpa'}
Subset: False
Superset: False

=== Code Execution Successful ===
```

15. Create a dictionary for a shopping list with items and quantities. Add 3 new items, update quantities for 2 items, delete 1 item using del and another m using del and another using pop(), print all keys, values, items, and check if a specific item exists.

shopping = {"milk": 2, "bread": 1, "eggs": 12}

shopping["butter"] = 1 shopping["cheese"] =

2 shopping["jam"] = 1 shopping["milk"] = 3

shopping["bread"] = 2 del shopping["jam"]

shopping.pop("butter") print("Keys:",

shopping.keys()) print("Values:",

shopping.values()) print("Items:",

shopping.items())

print("Is 'eggs' in shopping list?", "eggs" in shopping)

main.py | Output

```
Keys: dict_keys(['milk', 'bread', 'eggs', 'cheese'])
Values: dict_values([3, 2, 12, 2])
Items: dict_items([('milk', 3), ('bread', 2), ('eggs', 12), ('cheese', 2)])
Is 'eggs' in shopping list? True
```